
WHEN DO LINEAR-COST DIFFUSION PASSES HELP WITH LONG CONTEXTS?

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ABSTRACT

Linear-cost recurrent denoisers make repeated scans of long contexts affordable, but useful refinement also requires accurate conditional predictions. We use established learning/factorization-error accounting to distinguish these requirements and study a 455M-parameter bidirectional RWKV denoiser on exactly enumerable binary posterior tasks. A completed within-lineage intervention shows that balancing evidence positions during fixed-length training reduces far-context conditional error by 6.051 nats, with a conditional 95% interval [5.392, 6.735]. This result motivates a stronger prospective test: can source-calibrated errors select a reveal schedule on unseen dependency classes? The frozen experiment trains six fresh task-training lineages and compares policies with identical scan count and zero discarded dependence, sealing their predicted rankings before held-out evaluation. *Revision status: the new experiment is running; no transfer outcome or independent-lineage success is claimed in this working draft.* The earlier implementation-specific cost certificate remains supporting evidence, rather than an optimized deployment frontier.

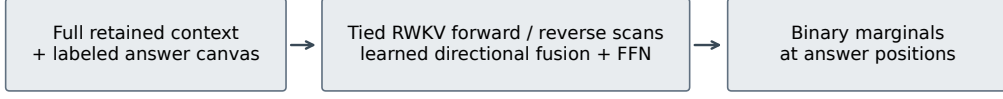
1 INTRODUCTION

Long-context generation must extract distant evidence and coordinate the answer tokens that depend on it. Linear attention and recurrent sequence mixers reduce the arithmetic required to process the input. Masked diffusion provides a separate mechanism: later calls condition unrevealed answer positions on tokens revealed earlier. Combining the two can make informative additional scans affordable, provided the learned conditionals remain accurate.

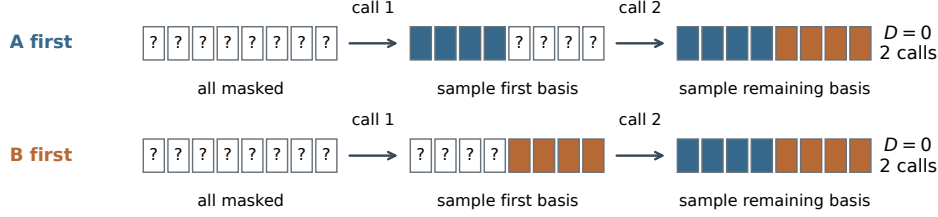
The usual factorization-error account explains why a suitable reveal schedule helps an oracle. It does not predict whether a trained model can use that schedule on a new context. An observed policy gain close to an oracle dependence gap is consequently a consistency check once both learned errors are small. Our scientific question is stronger: can measured conditional errors on source tasks predict which equally costly schedule will work better on unseen dependency structures?

We separate retrospective mechanism evidence from prospective prediction. The completed position intervention holds context length, examples and update count fixed; it supports improved distant-evidence use within one selected training lineage. The new experiment compares two public information bases of an invertible binary code. Both two-round policies have zero factorization loss, so their difference isolates learned error. A source-calibrated predictor makes its choices before target inference. Splitting whole code-invariant classes prevents the held-out set from being only a relabeling of training structures.

The intended contribution is an empirical test of transferable conditional-error prediction, grounded in established divergence accounting. Neither the KL identity nor recurrent diffusion is claimed as new. Six fresh task-training lineages address selection sensitivity for the new task, and all terminal outcomes are retained. The older quality–cost certificate is kept within its restricted comparison set; establishing a practical frontier would additionally require competent matched causal and optimized attention baselines.



Each call rebuilds recurrent states and reads the full context.



Public invertible bases: equal discarded dependence and scan count; learned errors may differ.

Figure 1: Implemented sampler and prospective comparison. Each call reads the full context and current answer canvas. Colored slots are sampled and retained; question marks are masked. The two public bases have equal oracle factorization loss and scan count. Actual prompts permute their labels and output positions.

2 SETTING AND IMPLEMENTATION

Let c contain H clamped context/query tokens and let $Y \in \{0, 1\}^N$ be the answer. The context is retained in full. A denoising call reads the context and the current answer canvas, then returns a binary marginal at each answer position. Previously revealed positions are preserved. For a fixed ordered partition $\pi = (B_1, \dots, B_K)$, positions in B_k are independently sampled using the model predictions conditioned on earlier reveals Y_{V_k} , where $V_k = \bigcup_{j < k} B_j$. The partition is chosen independently of the sampled values. This is the endpoint law audited below; confidence-ranked argmax decoding is a different law.

Recurrent denoiser. The main model adapts the released approximately 0.4B RWKV-7 backbone (Peng et al., 2025) into a tied bidirectional denoiser with 455,010,304 trainable parameters. Each call rebuilds transient recurrent states from the current canvas. It does not carry a scan’s terminal state into the next diffusion call. RWKV-7 is a generalized delta-rule recurrent mixer with linear sequence-length arithmetic at fixed width; it is not an exact implementation of softmax attention. We retain the distinction between recurrence implementing a chosen finite-feature kernel (Katharopoulos et al., 2020) and replacing that kernel by a learned generalized-delta operator.

At each recurrent block, the same mixer weights process the forward and reversed sequence. A learned directional gate fuses their outputs, followed by the feed-forward residual. Input features mark immutable, filled and masked positions and the denoising stage. The binary head is gathered only at answer slots. The model uses neither a hidden-depth loop nor a separate gate-modulation module. All model parameters are trained; the six new lineages share only the public pretrained backbone, not a task-adapted ancestor.

3 A COST-ERROR CRITERION

Fix c and write $P = P_c$. For each reveal group define

$$D_\pi(c) = \sum_k \mathbb{E}_{Y_{V_k} \sim P} \text{KL} \left(P(Y_{B_k} \mid Y_{V_k}, c) \parallel \prod_{i \in B_k} P(Y_i \mid Y_{V_k}, c) \right), \quad (1)$$

$$E_{M, \pi}(c) = \sum_k \sum_{i \in B_k} \mathbb{E}_{Y_{V_k} \sim P} \text{KL} (P(Y_i \mid Y_{V_k}, c) \parallel q_M^i(Y_i \mid Y_{V_k}, c)). \quad (2)$$

The first term is dependence discarded by independent within-group sampling; the second is error in the learned conditional marginals. With positive model probability on the target support, the chain rule gives

$$\text{KL}(P_c \| Q_{M,\pi,c}) = D_\pi(c) + E_{M,\pi}(c). \quad (3)$$

Conditional total correlation and factorization error are established tools for analyzing parallel discrete sampling (Lavenant & Zanella, 2025; Wen et al., 2026; Zhao & Cai, 2026). Equation (3) identifies the learned quantity to measure. Predicting that quantity before target evaluation is an additional empirical problem; the identity alone does not solve it.

Proposition 1 (Same-checkpoint intervention). *For two fixed reveal policies, multiple calls improve forward KL over one call if and only if*

$$E_{M,\text{multi}}(c) - E_{M,\text{one}}(c) < D_{\text{one}}(c) - D_{\text{multi}}(c). \quad (4)$$

Proof. Subtract the two instances of Eq. (3). $\square$

Supervision must cover the inference histories. The estimation term also depends on which histories receive supervision. Let μ_π select a target-distributed policy history and one of the N coordinate predictions uniformly, and let ν be normalized training supervision on the same context distribution. If $d\mu_\pi/d\nu \leq \kappa$, then $E_{M,\pi} \leq N\kappa \mathcal{R}_\nu$, where $\mathcal{R}_\nu$ is excess conditional cross-entropy above the oracle entropy floor. This standard change-of-measure bound does not hold with a finite uniform constant for uncovered histories. Appendix I gives a counterexample in which two-call excess error is zero while eight-call error is arbitrarily large. It also explains why raw minibatch loss is not a held-out competence certificate. For the implemented eight-stage absorbing objective, the exact worst-case coverage ratio is 1024 for either the information-set or sequential policy. Adding equal-weight information-set supervision reduces its own ratio below 2, while the sequential ratio remains about 1988. These are population risk-conversion constants, not sample-complexity lower bounds or guarantees that optimization attains a small risk.

Let $C_M(\pi, c)$ denote measured full-request cost in a specified unit. At fixed layer count and widths, recurrent mixing has $O(K(H + N))$ arithmetic, while dense softmax mixing has $O(K(H + N)^2)$ arithmetic. Projections, vocabulary heads, scan directions, precision, and hardware constants remain part of request cost. FlashAttention reduces memory traffic without making dense attention arithmetic linear (Dao et al., 2022). The asymptotic distinction motivates the experiment; it does not supply a measured speedup.

From accounting to a prospective decision. For two equal- D policies, the lower- E policy has lower endpoint KL. If a predictor satisfies $|E_\pi - \hat{E}_\pi| \leq \epsilon_\pi$ for each candidate, choosing the smaller predicted error has regret at most $\epsilon_{\text{chosen}} + \epsilon_{\text{optimal}}$. This standard stability observation follows by adding and subtracting the two predictions. It does not guarantee source-to-target calibration. The experiment below tests that empirical premise on contexts outside the source dependency classes. For supported decisions, let $n_\pi(z, c)$ count output predictions with public feature bin z and let $\hat{r}(z)$ be the source-calibrated error. The score is $\hat{E}_\pi(c) = \sum_z n_\pi(z, c) \hat{r}(z)$. Thus its policy contrast depends on the difference in feature counts. A common additive calibration offset cancels because each policy predicts eight outputs; feature-dependent residuals need not cancel. The held-out test asks whether these structural differences forecast the direction and useful magnitude of learned-error differences, rather than merely recovering the known D gap.

4 PROSPECTIVE EXPERIMENT

4.1 PREDICTING REVEAL SCHEDULES ON UNSEEN DEPENDENCY CLASSES

The fixed-partition decomposition identifies a prediction target but does not predict learned error on a new context. We test whether source-class calibration can supply that missing prediction. Let U be uniform on four binary variables and $P = BU \oplus s$, where B is invertible over GF(2) and s is public. Both U and P have independent fair marginals. Revealing either basis first therefore gives a two-round policy with $D = 0$. Both policies read the same full context twice. Their forward-KL difference is entirely a difference in learned conditional error, so dependence accounting alone cannot

rank them. All eight unconditional marginals are fair and the joint entropy is $4 \log 2$, so every one-call product sampler has forward KL at least $8 \log 2 - 4 \log 2 = 4 \log 2$.

We enumerate the 20,160 invertible matrices and partition their 25 homogeneous-code weight-enumerator classes by a fixed hash: 15 source classes and 10 held-out classes. Different weight enumerators certify that no permutation of the eight answer coordinates maps one code to the other. Equal enumerators need not imply equivalent codes. Training samples only source classes. Prompts randomize labels, coordinate order, equation order, and the names of the two publicly displayed bases. The selector obtains basis membership from the prompt rather than from private generator labels or target values.

Six independent task-training pipelines start from the same public pretrained backbone, using separate data and conditioning-initialization seeds. Each runs 300 two-bit updates, 400 four-bit updates, 1,800 source-code updates, and 600 16K updates with balanced public-block positions. Every terminal checkpoint is retained. This tests fresh lineages for the new task; it does not retroactively replicate the earlier position intervention across independent initial models.

For each model, source calibration comprises four fresh prompts in each source class at three context positions. We estimate per-coordinate conditional error using public features: fair versus deterministic marginal, parity fan-in, number of displayed equations combined (column degree for fair marginals), and position. A fixed binned estimator shrinks toward a round/position mean. A bin requires eight distinct source prompts; unsupported decisions fall back to a predeclared content-hash tie and remain in the primary denominator. First-round marginal error is included. No monotonic relation between parity fan-in and neural error is assumed.

Predicted scores and chosen policies for eight fresh prompts per held-out class, at all three positions, are hashed before any held-out neural evaluation. Exact evaluation then enumerates all 16 target endpoints under both policies. The primary estimand is KL reduction against the dependence-only content-hash tie. We also report fixed-basis, random-in-expectation, source-unconditional, and uncalibrated fan-in baselines, together with regret to the retrospective best policy. Uncertainty resamples training lineages and held-out code classes, then prompts within classes, retaining their paired positions. The interval conditions on the fixed source-calibration panel and shared pretrained backbone; it does not include uncertainty from choosing another calibration panel.

The predeclared success criteria require a mean improvement of at least 0.1 nat, a positive 95% interval, at least 80% supported decisions, six complete lineages, and at least five models whose mean selected-policy KL is below the one-call product floor $4 \log 2$. A gain over a dependence-only tie does not by itself establish that calibration improves on the structural heuristic. Cross-class error prediction is an empirical transfer hypothesis: the full-context training/test density ratio is infinite, so the preceding finite-coverage bound does not certify this extrapolation.

5 RESULTS AND LIMITS OF INFERENCE

Prospective test: outcomes pending. The six fixed-budget training pipelines have been launched after source-only runtime and numerical qualification. The source/test class split, calibration estimator, fallback, panels, estimands, success criteria and analysis source are frozen. All six prediction artifacts must be sealed before any held-out neural evaluation. This paragraph will be replaced by the complete lineage and held-out results. Training loss, runtime qualification, and algebraic feasibility are not evidence that the proposed predictor transfers.

Completed evidence motivating the test. In the earlier paired-parity study, three near/balanced adaptation pairs share one selected competent starting checkpoint. At fixed 16K training length, balanced positions reduce far conditional error by 6.0506 nats, with a problem-bootstrap interval [5.3916, 6.7353] conditional on those models. All three pairs favor balanced training. A fresh header/task layout panel retains a 7.6266-nat reduction [6.9081, 8.3015] with nearby instructions and distant task information. These interventions support evidence use under the declared layouts. They do not establish reliable full-pipeline learning or transfer to a different dependency class.

The exhaustive response audit retains three collateral errors among 55,296 unaffected comparisons. Regrouping distractors changes one model’s mean error by 4.09 nats. The earlier short-training

campaign succeeds only for seed71, and 1K-to-16K adaptation and changed-interface probes fail. These results constrain the mechanism interpretation and are retained in the supporting experiments.

Quality and cost answer different questions. The earlier equal-call gain agrees with $4 \log 2$ to within 1.3×10^{-4} nats when both learned errors are small. It does not independently forecast the error of an unseen policy or workload. The measured 1.30-second recurrent two-call versus 2.00-second attention two-call execution supports a restricted 1.5-second mean-budget certificate under its FP32/IEEE implementation. It does not establish an optimized practical frontier or a memory advantage. Cached causal decoding and other output factorizations remain outside that certificate (Appendix A).

Scope. The new task still has eight binary outputs, 16 valid endpoints, explicit public equations and synthetic distractors. Success would demonstrate prospective selection across the declared code classes and fresh task-training lineages, not open-ended language generation. All lineages share the same public pretrained backbone. Full context is retained between calls, so the mechanism does not recover information discarded by a state-only memory. Numerical checks use the FLA implementation; parity with official RWKV kernels remains unverified.

6 RELATED WORK

Masked discrete diffusion supplies the probabilistic construction (Austin et al., 2021; Sahoo et al., 2024). Linear Transformers separate a finite-feature kernel from its recurrent implementation (Katharopoulos et al., 2020); RWKV-7 develops the generalized delta-rule recurrence used here (Peng et al., 2025). DiffuMamba, Triplet-Block Diffusion RWKV, and retrofitted linear-attention diffusion already combine efficient sequence mixers with diffusion (Singh et al., 2025; Lin et al., 2026; Kim et al., 2026). We therefore do not claim the first recurrent diffusion language model. Analyses of factorized masked diffusion and conditional total correlation already address parallel sampling error (Lavenant & Zanella, 2025; Wen et al., 2026; Zhao & Cai, 2026). Information-theoretic discrete diffusion also connects denoising losses with mutual information and conditional likelihood (Jeon et al., 2025). The proposed revision tests whether source-calibrated conditional error can rank equal-cost schedules on unseen dependency classes. The completed positional intervention and its endpoint/counterfactual audits supply the motivating evidence.

REFERENCES

- J. Austin, D. D. Johnson, J. Ho, D. Tarlow, and R. van den Berg. *Structured Denoising Diffusion Models in Discrete State-Spaces*. NeurIPS, 2021. arXiv:2107.03006.
- T. Dao et al. *FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness*. NeurIPS, 2022. arXiv:2205.14135.
- A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret. *Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention*. ICML, 2020. arXiv:2006.16236.
- J. Kim, Y. Roh, and J. Kim. *Retrofitting Linear Attention into Diffusion Language Models*. arXiv:2608.06628, 2026.
- K. Lin, Y. Luo, Z. Su, Y. Song, and A. Rao. *Triplet-Block Diffusion RWKV*. arXiv:2605.25969, 2026.
- B. Peng et al. *RWKV-7 “Goose” with Expressive Dynamic State Evolution*. arXiv:2503.14456, 2025. Version 2, 30 March 2025.
- S. Sahoo et al. *Simple and Effective Masked Diffusion Language Models*. arXiv:2406.07524, 2024.
- V. Singh, O. Ostapenko, P.-A. Noël, E. Belilovsky, and T. Scholak. *DiffuMamba: High-Throughput Diffusion LMs with Mamba Backbone*. arXiv:2511.15927, 2025. Version 4, 17 July 2026.
- C. Wen, W. Liang, and J. Lu. *Conditional Total Correlation and the Serial Depth of Adaptive Parallel Sampling*. arXiv:2608.25505, 2026.
- Y. Zhao and C. Cai. *Adaptation to Intrinsic Dependence in Diffusion Language Models*. arXiv:2602.20126, 2026.

270 H. Lavenant and G. Zanella. *Error Bounds and Optimal Schedules for Masked Diffusions with Factorized*
271 *Approximations*. arXiv:2510.25544, 2025.
272
273 M. Jeon, S. Shin, D. Jeon, and A. No. *Information-Theoretic Discrete Diffusion*. arXiv:2510.24088, 2025.
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A RESTRICTED BUDGET CERTIFICATE

Proposition 2 (Budget certificate for a declared policy set). *Suppose a budget B permits recurrent model R 's multiple-call policy and attention model A 's one-call policy, but excludes all other policies in A 's declared comparison set. If*

$$D_{\text{multi}}(c) + E_{R,\text{multi}}(c) < D_{\text{one}}(c), \quad (5)$$

then the feasible recurrent policy has smaller forward KL than the feasible attention one-call product sampler.

Proof. The recurrent KL is the left-hand side. Every one-call product sampler has KL at least D_{one} , since its estimation error is nonnegative. $\square$

For a declared context law ρ , the same argument applies to $\mathbb{E}_\rho C_M$ and $\mathbb{E}_\rho$ KL by averaging both sides. Our empirical-panel check uses this mean-cost, mean-KL version; it supplies neither a per-request deadline nor pointwise quality guarantees.

The comparison is restricted to the declared policies and measured budget. It does not exclude more efficient untested attention implementations, correlated output heads, context-dependent or randomized mixtures of the declared policies, or a causal sampler. In particular, cached causal RWKV also admits a dependence-free sequential factorization and need not reread the prefix for each output token.

The missing long-context link. Increasing H does not itself change the dependence among N answer tokens. The learned error $E_{M,\pi}(H, \text{location})$ must be measured as evidence moves farther from the query. Bounds in generated length N cannot establish an advantage in clamped context length H . Our claim would be supported only where conditional competence and the measured cost conditions hold together.

B COMPLETED POSITION-INTERVENTION STUDY

B.1 PAIRED CONDITIONAL TASKS

Each public input specifies eight named output coordinates and four independent binary linear equations. Output slots retain their public coordinate labels. A base problem is paired across prefix lengths and public-block positions. The earlier study uses short, 1K, 4K, and 16K inputs; the matched study uses 16K primary inputs and short, 4K, and 32K secondary cells. Irrelevant archive records are inserted before and after the original instruction/evidence block, placing it near, midway, or far from the answer. The final answer footer remains next to the output. Actual token lengths, record counts, and block spans are recorded. Whole-record packing can undershoot the requested length slightly: the matched RWKV primary prefixes contain 16,383 tokens, while the attention cost prefixes contain 16,384 tokens under the same requested 16K budget. Reported context lengths count prefix tokens; public labels add a fixed $O(N)$ output scaffold. Moving instructions and equations together does not isolate factual retention from instruction retention.

The *systematic* control fixes four coordinates and leaves four independent fair bits, so $D_\pi = 0$ for every policy. The *paired-parity* family instead specifies four disjoint two-coordinate equations. After a public coordinate permutation, its uniform posterior has the form

$$Y = (U, U \text{ XOR } v), \quad U \sim \text{Uniform}(\{0, 1\}^4). \quad (6)$$

Both families have 16 valid outputs. For paired parity, one-call sampling loses $4 \log 2$ nats of dependence. Revealing an information set, then its complement, has zero within-group dependence error. The information set is computed from the public equation structure; it does not use a sampled answer. This deliberately controlled task tests the proposed mechanism rather than open-ended text generation.

Corollary 1 (Concrete learned-error threshold). *If the eight conditional marginal KL errors of the two-call sampler average at most $\epsilon(H)$, then*

$$\text{KL}(P_c \| Q_{R,\text{two},c}) \leq 8\epsilon(H), \quad Q_{R,\text{two},c}(\text{supp } P_c) \geq e^{-8\epsilon(H)}. \quad (7)$$

If $\epsilon(H) < \log(2)/2$, it has strictly smaller forward KL than every one-call product sampler on this target law.

Validity is reported alongside KL within the valid support: a concentrated sampler can be valid while failing the uniform target distribution.

A diagnostic for distant evidence use. On the paired task, write the information-set error as $E_{\text{info}} = E_{\text{first}} + E_{\text{second}}$. The first term measures four fair-marginal prediction errors; the second is the sum of four deterministic conditional cross-entropies, averaged over the 16 target-distributed first-round histories. Thus the stronger sufficient condition for beating every one-call product predictor is $E_{\text{second}}/4 < \log 2 - E_{\text{first}}/4$. This gives a specific learned quantity to improve through context adaptation.

A complementary local intervention flips one public parity constraint while holding the visible information-set values fixed. Exactly one remaining target flips. Orient its two logits so the required targets are respectively zero and one, and denote them by ℓ_0, ℓ_1 . With $\Delta = \ell_1 - \ell_0$ and $\text{sp}(x) = \log(1 + e^x)$,

$$L_{\text{flip}} = \frac{1}{2} \{ \text{sp}(\ell_0) + \text{sp}(-\ell_1) \} \geq \text{sp}(-\Delta/2). \quad (8)$$

Convexity proves the bound, with equality when $\ell_0 + \ell_1 = 0$. Small contrast imposes an error floor; a large common logit offset can still produce high error despite a large contrast. We therefore distinguish response strength from the nonnegative gap in Eq. (8), and check changes on unaffected coordinates. A fixed-history response probe is a local diagnostic; the exact endpoint audit supplies the full history-averaged error. Neither this elementary bound nor a response to one constraint alone certifies broad long-context competence.

B.2 INTERVENTIONS AND IMPLEMENTATION COSTS

The reveal-policy intervention holds the learned checkpoint and complete input fixed, changing only the reveal policy: one call, two information-set calls, or two calls that each reveal two intact dependent pairs. The two-call policies have equal group sizes. Let m_π count pairs revealed within the same group. Then $D_\pi = m_\pi \log 2$, and exact oracle marginal sampling has valid mass 2^{-m_π} . The intact-pair policy therefore has $D = 4 \log 2$ and oracle validity $1/16$, whereas the information-set policy has $D = 0$ and oracle validity one. Learned estimation error is measured for both policies; equal call count does not guarantee equal conditional competence. The observed contrast therefore contains both dependence and estimation terms; it is not an isolated causal measurement of dependence error alone.

Exact enumeration of the 16 target-support endpoints gives joint forward KL, valid mass, within-valid KL, and the D/E decomposition. Earlier short-task experiments also retain random partitions and sequential reveals (Appendix J). In particular, sequential factorization has $D = 0$ but can fail when its conditional histories are poorly learned.

The matched 16K cost check compares the recurrent denoiser with a 405M-parameter Pythia bidirectional softmax denoiser on the same public conditions. Native pretraining and tokenizers differ; at equal native-token budgets, irrelevant-record counts also differ. These are practical costs, not an architecture-causal quality ablation. An earlier separate-task screen also includes causal RWKV with prefix prefill and cached steps (Appendix G). Each batch-one request uses one H100; ranks process independent conditions. Full requests include every denoising call or the causal prefill, with warmup and CUDA synchronization. Shared audit caches are never used for timing. Cached and full causal predictions pass the declared numerical tolerance under IEEE arithmetic.

B.3 SELECTED TRAINING LINEAGE AND PRECEDING EVIDENCE

Only seed 71 in the earlier three-seed campaign learns the paired posterior; seeds 53 and 89 remain near chance. We select its fixed independent terminal checkpoint for the controlled mechanism study, retaining all six earlier outcomes. Both distance-intervention arms start from this same checkpoint. An earlier fixed 200-update adaptation at mostly 1K inputs improves performance at its trained length but fails the 16K aggregate. That completed study and all 640 evaluated conditions are retained in Appendix F. The completed position study controls training length at 16K and varies public-block position.

B.4 MATCHED 16K DISTANCE AND EVIDENCE INTERVENTIONS

The failure of 1K adaptation to extrapolate motivates a separately frozen 16K study. Both training arms start from the same selected seed-71 terminal checkpoint, with fresh optimizers and the original labelled answer interface. The near-only arm places the public block next to the answer; the distance-balanced arm distributes it equally across far, middle, and near positions. Both arms use 16K inputs and, within each adaptation seed, identical public examples, target-history draws, reveal policies, rounds, update counts, and optimizer settings. Only the block position differs. This controls training length and input-token budget. Instructions move with the equations, so the intervention concerns public-block distance rather than factual memory in isolation.

Both arms equally supervise the information-set and intact-pair two-call policies. Each samples a round uniformly and uses exact conditional soft targets for that round’s four coordinates. Half the public examples belong to each task family. The risk normalization and residual estimation-error bound are given in Appendix I; balanced supervision is not assumed to equalize learned errors. We freeze three adaptation seeds (20271011–20271013), global batch 32, AdamW learning rate 10^{-5} , 20-update warmup, weight decay 0.01, gradient clipping at one, and FP32/IEEE execution with TF32 disabled. A discarded six-update qualification sets a common fixed budget of 600 updates by a predeclared runtime rule. There is no quality-based early stopping or checkpoint selection. These are adaptation replicates from one selected lineage, not independent from-scratch training replicates.

The primary panel uses 32 fresh paired-parity problems at all three 16K positions, with exact endpoint laws for all three policies. Its primary contrast is near-only minus distance-balanced information-set conditional error at the far position. The original starting checkpoint is a descriptive baseline. Secondary panels use eight problems at the original short length, 4K at all three positions, and untrained 32K at far and near positions. All fixed checkpoints and cells are retained. The 32 public prompts are distinct from the preceding final and development panels, but use the same previously observed held-out structure catalog.

For every evaluated context, the counterfactual probe flips one declared parity constraint while holding all four visible information-set bits at zero. The native prefix changes at exactly one token, preserving token count and block position. The target of exactly one remaining coordinate reverses. We measure its signed logit contrast, mean correct conditional cross-entropy, the gap in Eq. (8), and probability changes on the three unaffected targets. This fixed-history intervention complements the endpoint audit, which integrates all 16 target histories.

Paired percentile intervals use 10,000 problem-bootstrap draws, preserving the pairing across arms, policies, positions, and adaptation seeds. Each seed’s effect is reported alongside the seed-averaged effect. These pointwise intervals condition on the three observed adaptation seeds; they do not estimate population training-seed uncertainty. The matched cost supplement uses the first eight predeclared problems at the three 16K positions, with one warmup and five complete timed requests per policy and model. It retains the 1.5-second mean budget and checks each fixed adapted model on this same eight-problem quality–cost panel. That secondary certificate is separate from the 32-problem primary mechanism analysis. The attention checkpoint supplies measured execution cost; its answer quality is not evaluated.

Distance coverage repairs conditional prediction at fixed length. All six 600-update adaptations and all 1008 evaluation conditions are complete. At 16K far, the three near-only models have information-set KL 8.3037, 7.0757, and 2.7725 nats; the matched balanced models have KL between 1.15×10^{-5} and 1.51×10^{-4} . The error reduction is positive for every seed (Table 1). Its seed-averaged value is 6.0506 nats, with a conditional problem-bootstrap 95% interval [5.3916, 6.7353]. The far-minus-near interaction is 6.0506 [5.3917, 6.7354], whereas the near-position difference is only -8.76×10^{-5} nats, with interval $[-2.77 \times 10^{-4}, 3.75 \times 10^{-5}]$. Both arms learn the nearby queries; the main difference concerns distant public blocks. These intervals condition on the three observed adaptation seeds and selected starting lineage, not a population of training runs.

The error reduction concerns the second pass and evidence use. For the original checkpoint, far-position KL is 21.8502 nats: first-round error is 0.00043 and second-round error is 21.84973. The matched training intervention reduces second-round error by 6.0505 nats on average, accounting for essentially the entire primary effect. The counterfactual probe distinguishes two failure patterns. Near-only seeds 20271011 and 20271012 have far evidence contrasts 0.791 and 0.608 but centering

Table 1: Primary 16K far-position contrast over 32 paired problems. Reduction is near-only minus balanced conditional error; here $D = 0$, so KL equals E . All three matched adaptation seeds are retained. Intervals resample problems, with the seed mean computed within each problem before resampling.

Adaptation seed	Near-only far KL	Balanced far KL	Reduction [95% CI]
20271011	8.3037	1.02×10^{-4}	8.3036 [7.0562, 9.6582]
20271012	7.0757	1.15×10^{-5}	7.0757 [6.0192, 8.1509]
20271013	2.7725	1.51×10^{-4}	2.7723 [2.7721, 2.7725]
Seed mean	—	—	6.0506 [5.3916, 6.7353]

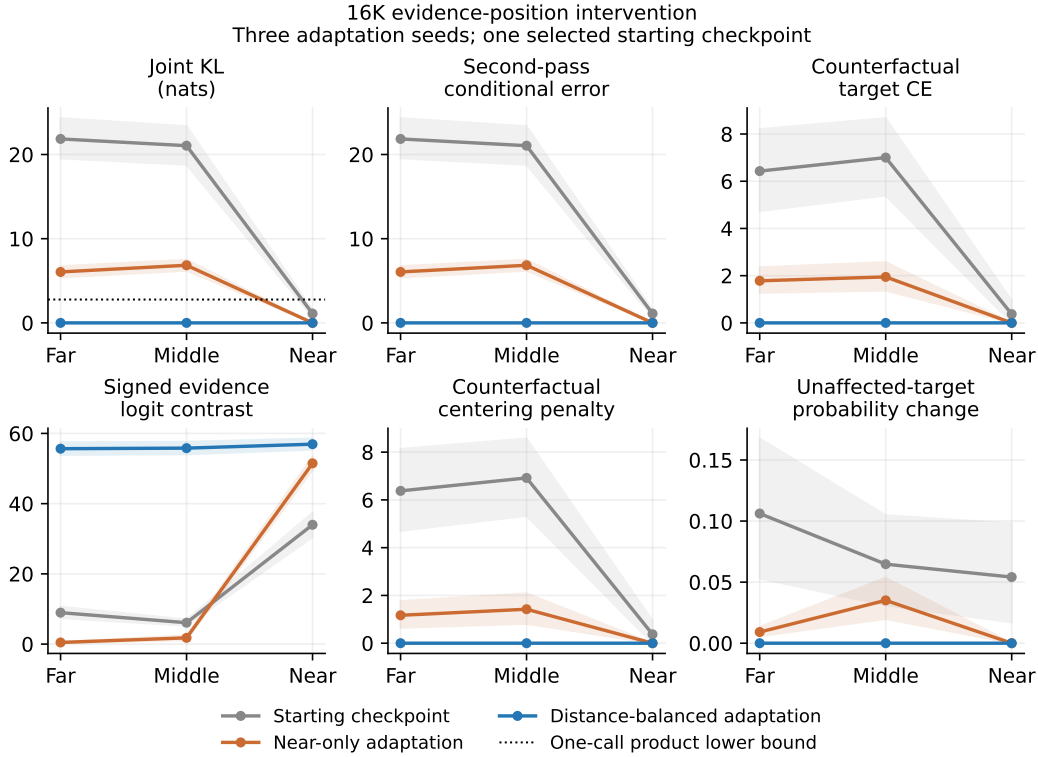


Figure 2: Matched 16K distance intervention. Curves average the three adaptation seeds within each of 32 paired problems; shading gives pointwise 95% problem-bootstrap intervals, conditional on the selected lineage and these seeds. The starting checkpoint is descriptive. The top-left dotted line is the one-call product lower bound. Endpoint error integrates all target histories; counterfactual panels use the declared fixed visible history. Numerically small balanced errors are shown explicitly in the tables.

penalties 1.591 and 1.915; seed 20271013 has contrast 0.000128 and CE 0.6931, close to an uninformative binary prediction. Balanced training raises the far contrast to 44.73–64.61 and lowers correct counterfactual CE below 3×10^{-7} in every seed. Mean changes on the three unaffected targets are below 2.7×10^{-6} for every balanced model. The seed-averaged counterfactual CE reduction is 1.7825 [1.2745, 2.3534]. Together with the stage decomposition, these selective responses support improved use of distant public constraints. They remain local fixed-history probes. An exhaustive follow-up over all histories and constraints supports the average error reduction but reveals rare collateral errors on unaffected targets (Appendix D); it does not establish uniform selectivity or identify a particular internal circuit.

Table 2: All fixed adaptations. The 16K KL and equal-call gain average 32 problems and three positions. Equal-call gain is intact-pair minus information-set KL; its dependence-only prediction is $4 \log 2$. Matched KL, latency, and certificate use the separate eight-problem, three-position panel and the 1.5-second mean budget. All KL values are nats.

Arm	Seed	16K KL	Equal-call gain	Matched KL	Two-call (s)	Certificate
Near-only	20271011	6.1875	-1.6543	5.2300	1.301	Fail
Balanced	20271011	1.99×10^{-4}	2.7725	6.31×10^{-4}	1.302	Pass
Near-only	20271012	4.8595	-2.0313	4.5207	1.307	Fail
Balanced	20271012	9.07×10^{-6}	2.7726	4.33×10^{-6}	1.306	Pass
Near-only	20271013	1.8482	0.9244	1.8483	1.301	Pass
Balanced	20271013	2.57×10^{-4}	2.7727	9.29×10^{-5}	1.302	Pass

The advantage survives nearby instructions and distant task information. A frozen follow-up separates the generic instruction header from the task block on 32 fresh prompts, testing four fixed-slot layouts and two legacy controls (Appendix E). With the header nearby and the task block far away, balanced training reduces conditional error by 7.6266 nats [6.9081, 8.3015], positive in every seed. Every balanced model’s mean error is below 0.000153 nats in that cell. Both arms remain accurate with nearby task information even when the header is distant. However, regrouping irrelevant records changes one near-trained model’s mean far error by 4.09 nats. The result supports the advantage under these declared layouts, not layout invariance. The task block includes names and output-order metadata; this inference intervention does not separate the components of the training treatment, and intervals still condition on the same selected lineage.

The learned equal-call contrast matches the dependence prediction. Across the three 16K positions, balanced information-set KL is 0.000199, 0.000009, and 0.000257 nats for the three seeds. The intact-pair minus information-set contrast is 2.772465, 2.772605, and 2.772696 nats. Its difference from the predicted $4 \log 2$ is at most 1.24×10^{-4} nats in absolute value. Thus the observed mean contrast is dominated by the prescribed dependence reduction once both conditional policies are competent. This conclusion requires the measured E values: near-only models have estimation-error differences from -4.804 to -1.848 nats, so their equal-call contrasts cannot be attributed to D alone (Table 2).

The matched budget certificate holds for all balanced selectors. On the separate eight-problem quality–cost panel, recurrent two-call times are 1.301–1.307 seconds across all six models. Attention costs 1.003 seconds for one call, 2.002 for information-set two-call execution, and 2.003 for intact-pair two-call execution. The predeclared 1.5-second mean budget therefore permits recurrent two-call and attention one-call execution while excluding both other declared attention policies. All balanced models have matched KL below $4 \log 2$, so all three meet the empirical sufficient certificate. Two near-only models fail its quality condition. The third passes the aggregate certificate because accurate nearby queries offset near-chance far and middle predictions; it still has far KL 2.7725. This distinction is why the primary mechanism test specifies the distant position instead of relying only on an aggregate score.

The certificate uses the analytic product–predictor lower bound, not measured attention answer quality. It supplies no per-request deadline guarantee and does not exclude optimized alternatives or cached causal decoding. Peak allocated inference memory is 4.125 GiB for RWKV and 2.234 GiB for attention; this implementation demonstrates no memory advantage. The exploratory eight-problem 32K far panel has balanced KL below 0.00041 in all three seeds, but it does not establish general extrapolation beyond the tested lengths. All secondary cells are retained in Appendix C.

C SECONDARY CELLS AND RESOURCE ACCOUNTING FOR THE MATCHED STUDY

Table 3 retains every secondary length/position cell for all seven fixed selectors. Each cell contains eight paired-parity problems. The 32K cells are exploratory tests at an untrained length. The original

checkpoint is descriptive, not a fourth adaptation replicate. The complete machine-readable results additionally retain every policy, endpoint law, evidence-flip probe, and timing request.

Table 3: Secondary information-set KL (nats). Seed suffixes 11, 12, 13 denote 20271011, 20271012, 20271013. No cell is removed by a competence gate. These eight-problem cells are separate from the 32-problem 16K primary panel.

Model	Short	4K far	4K middle	4K near	32K far	32K near
Original	0.0068	13.4858	14.0112	0.6453	45.6875	12.5611
near 11	0.0113	9.7235	10.0874	1.63×10^{-4}	11.9915	1.78×10^{-4}
balanced 11	9.84×10^{-5}	2.82×10^{-5}	2.31×10^{-5}	1.37×10^{-5}	4.06×10^{-4}	7.51×10^{-5}
near 12	1.06×10^{-6}	5.6200	4.7986	3.66×10^{-6}	14.3983	1.13×10^{-5}
balanced 12	3.29×10^{-6}	1.43×10^{-5}	1.39×10^{-5}	1.80×10^{-5}	1.30×10^{-4}	2.50×10^{-5}
near 13	4.62×10^{-6}	2.7745	2.7739	7.07×10^{-6}	2.7724	1.63×10^{-5}
balanced 13	5.38×10^{-5}	4.69×10^{-4}	7.17×10^{-4}	5.41×10^{-5}	6.40×10^{-5}	6.10×10^{-4}

Each main adaptation processes 315,283,200 native input tokens over 600 updates; the six matched runs therefore process 1,891,699,200 tokens. The discarded six-update qualification adds 3,152,832 tokens. Scheduler accounting for this follow-up totals 106.34 H100-hours, including qualification, all seven evaluations, the matched attention timing job, and three failed launches before model loading. It excludes the earlier checkpoint-training history described in the main text and preceding studies. Failed-launch receipts and resource-reservation/utilization details remain in the repository operations record. No failed attempt is removed from the cost.

Reproduction starts from the source and checkpoint hashes in the frozen training/evaluation manifests. The collectors validate raw endpoints, counterfactual token changes, numerical screens, and complete selectors before analysis. The paired report, figure, and matched certificate can then be regenerated with the report, figures, and cost-certificate scripts under `theory_mvp/distance_intervention/`. Publication tables copy the complete report; figure reflow changes typography only and checks that all plotted coordinates and uncertainty bands stay fixed.

D EXHAUSTIVE HISTORY AND CONSTRAINT RESPONSE AUDIT

The preceding intervention evaluated one syndrome flip at an all-zero visible history. After observing that completed study, we froze a separate diagnostic that enumerates all 16 target-distributed information-set histories and all four syndrome flips for every original primary problem, position and checkpoint. This is an exhaustive within-task diagnostic on reused problems, not an independent replication. No model is trained or selected in this follow-up. Each counterfactual changes one native input token and one required hidden target while preserving the visible history and complete input layout.

Connection to conditional error. For a fixed public problem x , let h index its 16 equiprobable visible histories, and let r index the four constraints. Each constraint controls one of the four remaining targets. Write c_{hr} for the correct target cross-entropy under the original prompt and c_{hr}^{flip} for the corresponding target cross-entropy after flipping constraint r . With $L_{hr} = (c_{hr} + c_{hr}^{\text{flip}})/2$, define

$$E_2(x) = \frac{1}{16} \sum_h \sum_r c_{hr}, \quad E_{2,\text{flip}}(x) = \frac{1}{16} \sum_h \sum_r c_{hr}^{\text{flip}}. \quad (9)$$

Direct summation and Eq. (8) give

$$4\bar{L}(x) = \frac{1}{2} \{E_2(x) + E_{2,\text{flip}}(x)\}, \quad \bar{L}(x) = \overline{\text{sp}(-\Delta/2)} + \bar{G}, \quad G \geq 0, \quad (10)$$

where bars average all 64 history–constraint pairs. The first identity follows because each hidden coordinate occurs exactly once in the constraint sum. The second averages the pointwise convexity decomposition. Crucially, $E_{2,\text{flip}}$ aggregates the intervened coordinate from four distinct counterfactual prompts; it is not the complete second-round error of any one prompt. Thus $\bar{L}$ measures

Table 4: Exhaustive response audit. $\bar{L}$ averages all histories and constraints; “both” is the fraction for which both evidence variants predict the required target correctly. $U = E_{\text{first}} + 8\bar{L}$ upper-bounds the original information-set error E . Last column is the worst paired CE over all 32 problems, 16 histories and four constraints, not the mean of per-problem maxima. All errors are nats.

Model	Position	$\bar{L}$	Both	E	U	Max CE
Original	far	5.256	0.06738	21.85	42.05	17.96
Original	middle	5.201	0.05469	21.05	41.61	17.1
Original	near	0.2027	0.9307	1.104	1.622	11.48
Near 11	far	1.99	0.04834	8.304	15.92	13.07
Near 11	middle	2.484	0.1255	10.26	19.87	13.66
Near 11	near	7.966e-05	1	0.000177	0.0006503	0.1119
Balanced 11	far	2.78e-05	1	0.000102	0.0002264	0.02725
Balanced 11	middle	0.0002888	0.9995	0.0004865	0.002317	0.3739
Balanced 11	near	8.537e-08	1	7.762e-06	8.244e-06	9.726e-05
Near 12	far	1.75	0.03955	7.076	14	13.57
Near 12	middle	1.856	0.09229	7.503	14.85	14.52
Near 12	near	1.157e-06	1	1.246e-06	9.951e-06	0.0008701
Balanced 12	far	1.214e-06	1	1.154e-05	1.404e-05	0.00178
Balanced 12	middle	8.617e-07	1	1.161e-05	1.231e-05	0.001531
Balanced 12	near	7.958e-09	1	4.058e-06	4.065e-06	1.189e-05
Near 13	far	0.6931	0.008789	2.772	5.545	0.6937
Near 13	middle	0.6931	0.02441	2.772	5.545	0.6942
Near 13	near	7.602e-09	1	6.989e-06	7.025e-06	2.549e-06
Balanced 13	far	2.973e-05	1	0.0001511	0.0002625	0.01098
Balanced 13	middle	4.269e-05	1	0.0001843	0.0003577	0.02056
Balanced 13	near	0.0001104	1	0.0004362	0.000905	0.06811

a symmetrized conditional-error law and must not replace the original endpoint E_2 . The original sufficient criterion remains $E_{\text{first}} + E_2 < 4 \log 2$. Nonnegativity of $E_{2,\text{flip}}$ also gives a conservative sufficient criterion directly from the exhaustive response diagnostic:

$$E_{\text{first}} + 8\bar{L} < 4 \log 2 \implies \text{KL}_{\text{info}} < 4 \log 2. \quad (11)$$

This follows by substituting $E_2 \leq 8\bar{L}$; it is an algebraic consequence of the measured target law, not a training-to-test guarantee. The response-based criterion is sufficient but not necessary, so failing it does not establish failure of the exact endpoint criterion.

Analysis and checks. The primary contrast is near-only minus balanced mean paired CE at 16K far, with equal averaging over the 16 histories and four constraints. We report each adaptation seed and their mean, using 10,000 paired problem bootstrap draws with seed 20260923. Histories and constraints are exact inner expectations, not independent statistical samples. Intervals remain conditional on the observed checkpoints and selected lineage. All seven selectors and all three positions are retained. We separately report the response floor, centering penalty, fraction of pairs with both target predictions correct, unaffected-target probability changes and worst-case errors. Mean improvement alone does not establish uniform competence. The original first/second errors and original fixed-history probes are recomputed, with a predeclared absolute reconciliation tolerance of 10^{-4} for each compared metric. Raw normalized binary probabilities are retained for every call.

Exhaustive results. The audit retains all 672 conditions and 43,008 paired interventions. The far near-only minus balanced paired-CE contrast is 1.4779 nats (conditional 95% interval [1.3275, 1.6371]). The three individual seed contrasts are 1.9903, 1.7501, 0.6931 nats. Recomputed archived metrics agree to within 8.3×10^{-5} in absolute value. Table 4 reports every checkpoint and placement. The response-based upper bound is an additional algebraic diagnostic; the frozen primary analysis uses the paired-CE contrast. Every balanced checkpoint satisfies the conservative response-based criterion for every one of the 32 problems at every measured position. This statement is limited to the enumerated problem panel. The largest unaffected-target probability change among all balanced interventions is 0.992. This worst-case value is reported separately from average selectivity.

Rare collateral failures. A post hoc inspection of these large shifts finds 3 correct-to-incorrect changes among 55,296 unaffected-target comparisons across the three balanced models, 32 problems, all positions, histories and constraints. These are dependent comparisons within the enumerated panel, not independent test examples. Averaging complete second-round errors on the flipped prompts, including unaffected targets, gives at most 0.002384 nats for any balanced model/position cell. The first-round flipped-prompt error was not measured, so this is not a full counterfactual endpoint KL. The findings support low average conditional error and mostly selective responses, while ruling out a claim of uniform selectivity. The conservative certificate in Eq. (11) concerns the original target law and does not certify the complete counterfactual prompt.

Additional resources. The seven evaluations consume 11.89 additional H100-hours, with no retraining or failed launches. Detailed resource and utilization records remain in the repository.

E SEPARATING INSTRUCTION-HEADER AND TASK-BLOCK POSITIONS

The training-distance intervention moves instructions and task information together. A separately frozen inference experiment tests whether the balanced models retain their advantage when the generic instruction header stays nearby and the task-bearing block moves far away. This intervenes on inference layout; it does not separately identify the contributions of instruction and task distance in the preceding training treatment.

Independent fixed slots and fresh problems. We draw 32 paired-parity problems with seed 2026092301 and use all six fixed adaptation checkpoints. These public prompts have no overlap with the previous 32-problem panel, but use the previously observed held-out structure catalog. The 24-token generic instruction header and 88-token task block move between independent far and near slots. The latter block contains coordinate names, equations and output-order metadata; this is not an intervention on syndrome distance alone. The native prefix has 16,383 tokens and the answer footer remains fixed. Unoccupied slots contain deterministic groups of whole archive records with exactly matching token lengths. Swapping a block with its filler preserves the complete token multiset and the other block’s exact position. All four header/task-position combinations are tested. Two additional legacy far/near layouts retain the original serializer on the same new problems, measuring any effect of regrouping irrelevant records in the fixed-slot design.

Prediction and analysis. Position does not change the target dependence term: $D_{\text{info}} = 0$. Thus, with h the header position and t the task position,

$$\text{KL}_{\text{info}}(h, t) = E(h, t), \quad P(h) = E(h, \text{far}) - E(h, \text{near}). \quad (12)$$

The primary contrast is near-trained minus balanced-trained error at $(h, t) = (\text{near}, \text{far})$. We additionally report the task-distance penalty $P(h)$ at each header position, header penalties at each task position, their factorial interaction, training-arm differences and legacy discrepancies. The predeclared stronger interpretation requires a positive primary interval, positive direction in all three matched seeds, and balanced mean error below $4 \log 2$ in both nearby-header cells for every seed. A positive arm contrast alone need not yield a useful two-pass predictor.

All 16 target histories and one-call/information-set endpoint laws are evaluated for every condition. Ten thousand paired problem-bootstrap draws use seed 2026092302, preserving pairing across models and layouts. Intervals remain conditional on these six checkpoints and the selected starting lineage. The standard numerical screen is supplemented by repeated/cross-rank checks of all four split layouts with both all-masked and visible-history inputs. Every cell and control is retained regardless of accuracy.

Results. All 1,152 conditions and 2,304 endpoint laws are validated. The primary error reduction is 7.6266 nats, with conditional 95% interval [6.9081, 8.3015]. The individual adaptation-seed effects are 12.4718, 7.6350, 2.7730 nats. The predeclared stronger-interpretation criteria are satisfied. Table 5 includes every layout and both legacy controls. Reduction in task-distance penalty with nearby header: 7.6266 nats [6.9081, 8.3015]. Reduction in task-distance penalty with distant header: 8.6760 nats [7.7375, 9.6152]. The largest absolute mean co-located-versus-legacy discrepancy is 4.09 nats; these controls quantify the additional distractor regrouping.

Table 5: Information-set KL for independent header/task positions, averaged over 32 fresh problems. FF, FN, NF and NN denote header then task position (far or near). Legacy controls use the original whole-block layouts. Seed suffixes identify the same six fixed adaptations used throughout. All entries are nats.

Model	FF	FN	NF (primary)	NN	Legacy far	Legacy near
Near 11	13.63	0.000134	12.47	6.82e-05	9.544	6.704e-05
Balanced 11	1.235e-05	7.123e-06	1.344e-05	8.588e-06	4.081e-05	8.635e-06
Near 12	9.621	1.706e-06	7.635	1.017e-06	7.282	1.03e-06
Balanced 12	3.114e-06	3.535e-06	3.845e-06	4.145e-06	4.248e-06	4.131e-06
Near 13	2.773	8.744e-06	2.773	7.057e-06	2.773	7.17e-06
Balanced 13	9.141e-05	6.024e-05	0.0001524	0.0001064	8.012e-05	0.0001063

Table 6: Complete factorial contrasts within each checkpoint. P_F, P_N are task-distance penalties with far/near header; H_F, H_N are header-distance penalties with far/near task. $I = P_F - P_N$. All are paired mean KL differences in nats.

Model	P_F	P_N	H_F	H_N	I
Near 11	13.63	12.47	1.162	6.583e-05	1.162
Balanced 11	5.229e-06	4.854e-06	-1.09e-06	-1.465e-06	3.743e-07
Near 12	9.621	7.635	1.986	6.886e-07	1.986
Balanced 12	-4.214e-07	-2.999e-07	-7.315e-07	-6.1e-07	-1.216e-07
Near 13	2.773	2.773	0.0002761	1.687e-06	0.0002744
Balanced 13	3.117e-05	4.6e-05	-6.103e-05	-4.62e-05	-1.483e-05

Scope and resources. This test uses fresh prompts within the existing catalog and existing training lineage. The task-bearing block includes mapping and output-order information, so the contrast does not isolate factual retention from every other task requirement. No new latency certificate, independent training replication or natural-language result is claimed. The six completed evaluations and one stopped startup attempt use 6.60 additional H100-hours. The stopped attempt produced no predictions and was manually replaced using the same frozen evaluation. Its zero reported runtime and startup logs are both retained in the operations record.

What the factorial test identifies. Because D is fixed, these layout contrasts are changes in learned conditional error, not changes in the target’s dependence structure. The positive primary contrast shows that the training-arm advantage survives with nearby generic instructions; it does not imply that header position has no effect. Fixed token multisets control content and length, but moving blocks necessarily changes sequence order. In particular, the co-located versus legacy contrasts measure sensitivity to regrouping irrelevant records even with the same meaningful-block positions. We therefore interpret the result as a test of competence under the declared layouts, not proof of a unique memory or retrieval circuit. It neither separates all components of the earlier training treatment nor adds a new architecture-level cost comparison.

F EARLIER 1K ADAPTATION AND ITS 16K EXTRAPOLATION FAILURE

This completed experiment precedes the matched 16K distance intervention in Section B.4. Its evaluation seed, prompts, adaptation budget, and timing panel differ. Its numerical results are retained separately and are not substituted into the later matched quality–cost comparison.

F.1 EARLIER SELECTION AND FROZEN EVALUATION

The earlier three-seed campaign learns the paired posterior only for seed 71; seeds 53 and 89 remain near chance. We therefore explicitly select the fixed seed-71 independent terminal checkpoint for a conditional mechanism study, while retaining all six outcomes. A fresh eight-condition development probe first checks short competence, then measures every long-context cell. The one bounded follow-up preserves the task and labelled output format: 200 fixed updates at global batch 32, learning rate 10^{-5} with 20-update warmup, using 25% short and 75% 1K inputs and the original independent

Table 7: Predeclared 16K paired-parity aggregate, equally weighting positions within each of 16 problems. KL gain is one-call minus two-call KL; positive is favorable. Intervals concern problems, not training seeds.

Checkpoint	One-call KL	Two-call KL	Two-call validity	KL gain	95% interval
Before adaptation	2.797	14.106	0.319	-11.309	$[-13.620, -8.873]$
After adaptation	2.773	15.496	0.237	-12.723	$[-15.808, -9.968]$

paired information-set objective. Its discarded 20-update GPU qualification passes on all eight ranks. All 2,500 auxiliary training updates remain part of the model’s history; adaptation starts with a fresh optimizer.

Before evaluating these two fixed checkpoints, we freeze 16 base problems per family, test seed 20271001, all ten length/position cells, and both fixed checkpoint selectors: the original model and its terminal 200-update adaptation. Every cell is retained regardless of short-task performance. The test structure catalog was used in the earlier campaign, so this is a fresh-condition assessment within a previously observed held-out catalog, conditional on a selected training lineage. It is not an untouched-structure test or evidence about reliability across training seeds.

The earlier study’s primary estimand is information-set improvement over one call in paired parity at 16K, averaging the three evidence positions within each base problem. We also report the equal-call contrast, valid mass, all cells, and each gain’s interaction with short context. Pointwise 95% percentile intervals use 10,000 bootstrap draws over base problems, retaining all policies, lengths, positions, and both checkpoints together. No endpoints are discarded and no sample extension depends on significance. All 640 conditions across the two checkpoints are complete and validated. Exact public prompts have no overlap with the earlier confirmation panel or the eight-condition development probe.

F.2 RESULTS OF THE EARLIER ADAPTATION

Learning at the trained length does not ensure extrapolation. Before context adaptation, paired-parity information-set KL is 0.0407 nats on the fresh short-input panel. At 1K, it rises to 28.53 and 29.66 for far and middle public blocks, while remaining 0.383 for nearby blocks. The fixed 200-update adaptation reduces these 1K errors to 0.0127, 0.0185, and 0.0121, with valid mass 0.9922, 0.9897, and 0.9928. Thus the model can learn useful conditional predictions at the trained length.

At 4K, adapted information-set KL is 1.842, 0.992, and 0.148 for far, middle, and near placements. The mean gains over one call are positive, but the far-position gain interval includes zero: 0.931 nats with interval $[-0.587, 2.046]$. At 16K, information-set KL rises to 21.83, 21.41, and 3.245. Figure 3 retains every length and position. Improving in-distribution conditional predictions does not imply that the same error budget holds at a longer, untrained context length.

The primary 16K criterion fails. Averaging evidence positions within each problem, two calls worsen KL relative to one call by 11.309 nats before adaptation and 12.723 afterward (Table 7). Both paired intervals exclude zero in the unfavorable direction. The adaptation’s change in two-call KL itself is inconclusive at this aggregate: KL reduction -1.390 , interval $[-5.448, 2.678]$. We do not interpret a larger point estimate as a reliable overall degradation.

Validity alone gives a different impression. Two calls increase average valid mass over one call by 0.257 before adaptation and 0.175 afterward, despite the worse forward KL. A model can put appreciable mass on valid answers while strongly distorting their uniform target distribution. Reporting within-valid KL and full endpoint KL prevents this from being mistaken for a better posterior sampler.

The equal-call intact-pair control also illustrates the role of estimation. After adaptation, the information-set policy improves mean 16K KL over that control by 18.685 nats, interval $[16.326, 21.392]$. Only $4 \log 2 = 2.773$ nats of this contrast is the prescribed dependence reduction; the remaining 15.912 nats is the difference in learned estimation errors. Thus the same-call intervention does not isolate dependence error alone. The successful information-set histories and poorly

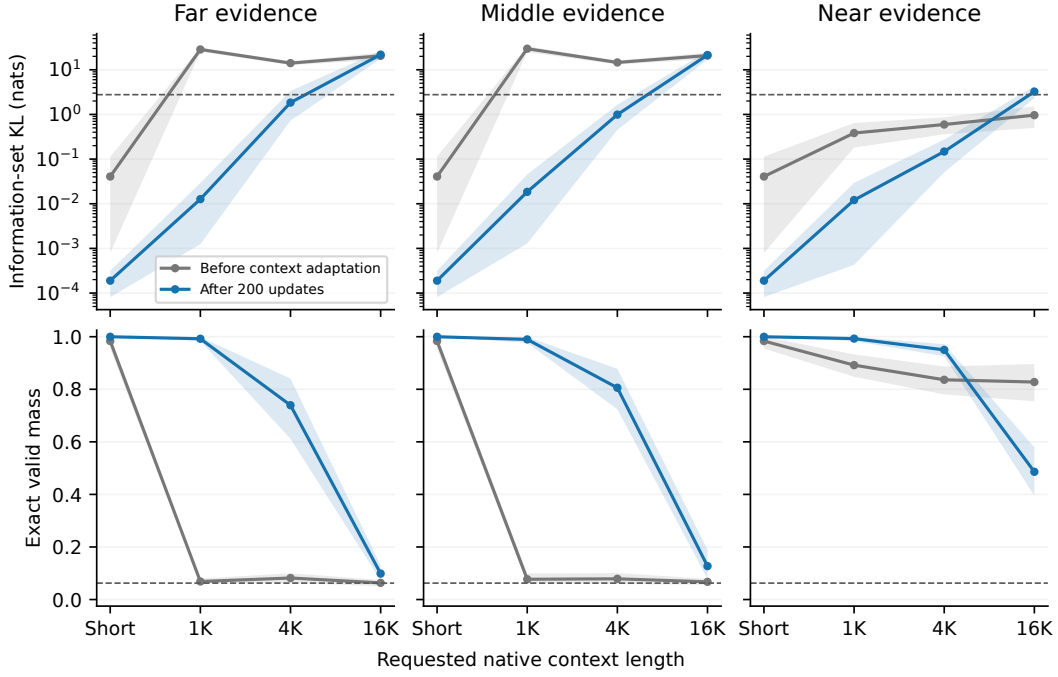


Figure 3: Frozen fresh-condition results for the information-set policy. Each cell averages 16 paired-parity problems; bands are pointwise 95% problem-bootstrap intervals. The dashed KL line is the one-call product lower bound $4 \log 2$, and the dashed validity line is $1/16$. Both checkpoints share the selected seed-71 training lineage.

learned alternative histories are consistent with the coverage analysis, without proving a particular internal cause.

Matched-task cost conditions hold; the quality certificate does not. Before collecting these timings or observing the adapted final outcomes, we fix a 1.5-second mean-request budget. The cost check uses the same 16 paired-parity problems and three 16K placements, with one separate warmup and ten complete timed requests per condition and policy. The recurrent checkpoint is the fixed 200-update adaptation; the Pythia adapter is its previously qualified 20-update checkpoint. Attention accuracy is not measured: the comparison uses the analytic lower bound for any one-call product sampler.

Mean recurrent latencies are 0.650 seconds for one call and 1.300 seconds for the information-set two-call policy. Attention takes 1.000 and 1.998 seconds, respectively; its intact-pair two-call policy takes 1.999 seconds. The pointwise problem-bootstrap intervals for the information-set costs are $[1.298, 1.302]$ and $[1.993, 2.004]$ seconds. Hence the predeclared budget permits recurrent two-call and attention one-call execution while excluding both other declared attention policies. Nevertheless recurrent mean KL is 15.496, well above the $4 \log 2 = 2.773$ sufficient threshold. The empirical-panel certificate fails its quality condition. This failure does not determine the unmeasured attention model’s actual KL.

These are mean-cost measurements, not per-request deadline or broad population guarantees. Peak allocated memory is 4.125 GiB for recurrent two-call execution and 2.234 GiB for attention; this implementation establishes no memory advantage. The earlier auxiliary screen finds attention faster at 1K and 4K and cached causal RWKV cheaper at 16K (Appendix G); a linear arithmetic count alone does not imply universal speed superiority.

G AUXILIARY LOOKUP ADAPTATION AND COST INPUTS

A separate development task supplies key-value records and a query key, with disjoint train/development/test key banks. Retrieval asks for the queried four-bit value repeated twice; the dependent task asks for $(U, U \text{ XOR } v)$. The 600-update recipe uses global batch 32, 200 short-context updates followed by 400 at 1K, with equal ordinary-denoising and two-round-history objectives. Starting from the released checkpoint, information-set validity at update 600 is 0.999964 for retrieval and 0.062652 for dependence (KL 0.000036 and 2.788015 nats). The archived update-200 model already shows the same separation, with validity 0.999997 and 0.063186.

Starting this adaptation from the competent seed-71 posterior checkpoint also fails its short gate: at update 600, retrieval validity is 0.562927 and dependent validity is 0.092507 (KL 0.678439 and 2.607604). This changes both public syntax and labelled output slots to a contiguous binary suffix, so it does not isolate initialization or context length. Both fixed outcomes are retained; no long accuracy sweep or full baseline task training follows their failed gates. These results motivated preserving the learned interface in the earlier context probe.

The separate cost screen uses the released-initialization 600-update RWKV checkpoint and discarded 20-update qualification checkpoints for the Pythia and causal RWKV adapters. It retains every declared input/policy cost even though the accuracy gates fail. Precision is FP32/IEEE with TF32 disabled; all long-length numerical checks pass. These timing results do not establish a joint quality-cost certificate for the changed labelled-equation panel. The separate-task mean two-call times at 1K/4K/16K are 0.279/0.417/1.302 seconds for recurrent execution and 0.056/0.217/2.000 seconds for attention. Cached causal RWKV takes 0.416/0.445/0.729 seconds. The earlier matched-task 16K check in Appendix F therefore supplies the relevant cost evidence for its own labelled inputs.

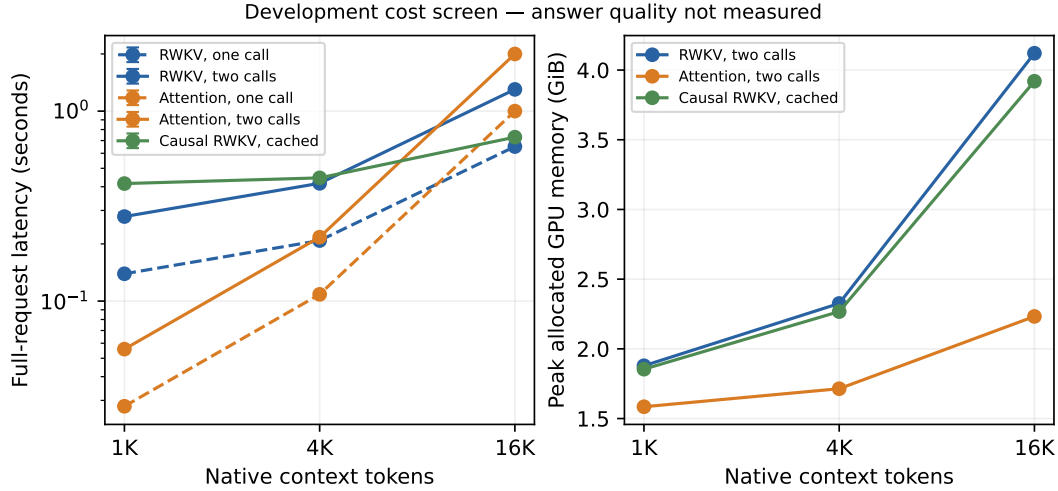


Figure 4: Supporting lookup-task request-cost screen. Whiskers span condition means, not confidence intervals. Native tokenizers differ. This figure measures no answer quality and is separate from the matched-task main-text check.

H EXACT ENDPOINT ACCOUNTING

For a fixed partition the target factorizes by groups, while the model factorizes within groups:

$$P(y \mid c) = \prod_k P(y_{B_k} \mid y_{V_k}, c), \quad Q(y \mid c) = \prod_k \prod_{i \in B_k} q^i(y_i \mid y_{V_k}, c).$$

Insert and subtract $\sum_{k, i \in B_k} \log P(y_i \mid y_{V_k}, c)$ inside $\mathbb{E}_P \log(P/Q)$. The two resulting terms are exactly D_π and $E_{M, \pi}$. The averages use target histories, not model rollout histories. Enumerating valid target support therefore suffices for forward KL; it does not estimate the probability of invalid paths by omitting them. Their total mass is obtained by subtracting the exactly summed valid endpoint probability from one.

For valid support S and $V = Q(S) > 0$,

$$\text{KL}(P\|Q) = -\log V + \text{KL}(P\|Q(\cdot | S)).$$

Nonnegativity gives $V \geq e^{-\text{KL}(P\|Q)}$. For the dependent task the first four bits are independent and the second group is deterministic given the first and the public value; hence $D_{\text{two}} = 0$. Each individual bit is fair and the answer entropy is $4 \log 2$, so $D_{\text{one}} = 8 \log 2 - 4 \log 2$. This proves the concrete threshold and validity certificate.

For a general partition of the four independent pairs, a pair revealed in different groups is sampled as one fair bit followed by a deterministic conditional bit, contributing zero dependence error. Oracle conditionals on impossible target histories may be extended arbitrarily: once a committed history violates a completed pair, it cannot lead to a valid endpoint. A pair revealed within one group is sampled as two fair bits and contributes $\log 2$, with probability $1/2$ of satisfying its equation. Independence across pairs proves $D_\pi = m_\pi \log 2$ and oracle validity 2^{-m_π} . Among the 70 uniformly selected four/four partitions, the counts for $m_\pi = 0, 2, 4$ are 16, 48, 6. Consequently expected dependence is $(12/7) \log 2$ and expected oracle validity is $227/560$. These are analytic predictions; they are not learned-model measurements or a substitute for the fixed realized partitions in the audit.

I HISTORY COVERAGE AND CONDITIONAL EXCESS RISK

A prediction query z includes the public context, visible coordinates and values, target coordinate, and model stage input. Write $e(z)$ for its conditional marginal KL. Then $E_{M,\pi} = N \mathbb{E}_{\mu_\pi} e(z)$. If $\mu_\pi \ll \nu$ with density ratio at most κ ,

$$E_{M,\pi} = N \mathbb{E}_\nu \left[\frac{d\mu_\pi}{d\nu} e(z) \right] \leq N \kappa \mathbb{E}_\nu e(z).$$

The expectation on the right is supervised cross-entropy minus the exact conditional-entropy floor. Both sides must use the same context law; this is not a generalization bound from training keys to test keys or longer contexts. For balanced two-round supervision on the eight-bit task, population excess risk is $E_{M,\text{info}}/8$. If this objective has mixture weight α , nonnegativity of the other objective's excess risk gives $E_{M,\text{info}} \leq (8/\alpha) \mathcal{R}_{\text{mixture}}$. A family-specific guarantee requires its conditional context law, rather than using the combined-family risk without adjusting for the sampling probability.

For the necessity of coverage, set predictions to the exact oracle on the empty and first-four-visible histories used by the information-set policy. On its unvisited one-bit-visible history, give the next, still-fair bit probability ε of being one. Leave other predictions at the oracle. The two-call KL is zero, but sequential KL includes

$$\text{KL}(\text{Bern}(1/2) \| \text{Bern}(\varepsilon)) = -\log 2 - \frac{1}{2} \log[\varepsilon(1-\varepsilon)] \longrightarrow \infty.$$

Both policies have $D = 0$. Ordinary denoising can supply coverage, but rare histories may leave a large density ratio. The observed seed-71 policy mismatch is compatible with this argument; it does not identify its internal causal mechanism or establish a long-context effect.

Exact mask-coverage constants. Generalize the implemented objective to N uniformly sampled stages, masking probability $p = t/N$, and weight $1/t$ per masked coordinate. For a particular mask M and $i \in M$, the normalized supervision weight is $p^{|M|}(1-p)^{N-|M|}/(Nt)$. A policy query has weight $1/N$ and uses stage $t = |M|$, giving density ratio

$$\frac{t}{(t/N)^t(1-t/N)^{N-t}}.$$

For an even- N balanced two-round policy, the sharp maximum is $(N/2)2^N$, attained in round two; round one has ratio N . At $N = 8$, equal mixing with information-set supervision yields $\kappa_{\text{info}} = 2048/1025 < 2$, but $\kappa_{\text{sequential}} \approx 1988.411$ for the canonical first-four information set (and 2048 for a last-four information set), while $\kappa_{\text{pair-preserving}} = 2048$. Enumeration of all masks verifies these constants. The corresponding bound on total estimation error is $N\kappa\mathcal{R}_\nu$. This quantifies a limitation of changing inference policies after training. It is a worst-case coverage statement about answer length N , not an actual sample-complexity lower bound, an explanation proved for a particular trained network, or an advantage in context length H .

Balanced supervision of both two-call policies. An equal mixture of information-set and intact-pair supervision samples a round uniformly and uses $(2/8)$ times the sum of its four coordinate losses. For a common public-context law, its population excess risk is therefore

$$\mathcal{R}_{\text{mix}} = \frac{E_{\text{info}} + E_{\text{pair}}}{16}.$$

The paired-task oracle entropy floor is $(3/4) \log 2$; averaging equally with the systematic family gives $(5/8) \log 2$. Random-round minibatches can have different floors, which must be computed from their actual conditional targets. On paired parity, nonnegativity yields

$$|\text{KL}(P \| Q_{\text{pair}}) - \text{KL}(P \| Q_{\text{info}}) - 4 \log 2| \leq 16 \mathcal{R}_{\text{mix}}.$$

Thus a small held-out risk bounds the estimation-error contribution to the same-call contrast. Equal supervision weights alone do not establish equal competence. Also $\mathcal{R}_{\text{mix}} < \log 2/4$ suffices for the information-set policy to beat every one-call product predictor on this law. If risk additionally averages both families and three positions equally, the corresponding constants for paired-family mean-position error and specified-position error are respectively 32 and 96. All statements require the same underlying public-example distribution; they do not convert training minibatch loss into held-out or longer-context guarantees. Near-only supervision leaves far-position public queries uncovered.

J EARLIER CHECKPOINT AND POSTERIOR EVIDENCE

The frozen F2 checkpoint study contains 52,800 accepted synthetic examples per model across supplied lengths 16K, 32K, and 64K. F2 and released causal RWKV obtain 15 and 14 exact matches. Conditional macro accuracies are 0.030864% and 0.028807%; their difference is 0.002058 percentage points, with a paired conditional 95% bootstrap interval of $[-0.020576, 0.024691]$. Sixty unsupported cells are exclusions. Evidence-only controls yield zero correct and zero parseable answers out of 72 per model. These outcomes do not isolate retention.

A separate 455M development study compares independent and complementary policy histories from the same seed-17 parent. At update 2,500, information-set parity KL is 2.800646 and 2.772621 nats, with valid mass 0.062189 and 0.062500: both remain near the 1/16 baseline. The frozen three-seed replication retains all six terminal branches (Table 8). Only seed 71 learns the supervised information-set law. Its independent and complementary branches have eight-call sequential KL 11.447909 and 14.174383 nats: additional calls alone do not deliver the learned law. All seed-53 and seed-89 branches remain near chance. These results expose training-seed and policy-history sensitivity; a successful seed does not establish reliable learning. The lookup task changes the input/output interface and is reported separately (Appendix G).

Table 8: All fixed-terminal short-task replication branches: information-set paired-parity endpoints on the frozen 54-condition panel.

Seed	Independent KL	Valid mass	Complementary KL	Valid mass
53	2.773932	0.062501	2.772756	0.062500
71	0.007372	0.996466	0.101755	0.973031
89	2.773184	0.062499	2.772615	0.062500

Full checkpoint panels, training histories and numerical qualifications remain in the preserved experiment records. These earlier panels do not identify the position intervention summarized in the main text.

K MODEL ACCESS AND REPRODUCTION

The supplementary release provides lossless inference exports of the selected initial seed-71 model and all six near/balanced adaptations used in the completed studies. Every original model tensor and checkpoint contract is retained; optimizer state is omitted. The registry records both the original checkpoint hash and the new export-container hash, with verified tensor-content identity. A supplied wrapper checks these identities and rebinds copied evaluation manifests without changing scientific source code or task panels. Evaluation still requires the recorded eight-H100 numerical environment. Public model access and tensor identity do not constitute independent neural replication.

A separate reconstruction workflow starts from pinned public base weights and includes the original parent, independent-history branch and all six adaptations. The prospective study has its own exact training/evaluation source capsules and complete portable commands through calibration, prediction sealing, held-out evaluation and independent raw-data analysis. Synthetic integration checks exercise the complete analysis and compressed-evidence workflow, including retention of chance-level models and failed competence gates; these checks are software validation and are not additional experimental outcomes.